Feynmann Excercises Physics

October 28, 2026

1 Atoms in Motion3

Refer to chapters 1-3.

1.1 If heat is merely molecular motion, what is the difference between a hot , stationary baseball and a cool, rapidly moving one?

- A hot stationary baseball is made of atoms that are moving and bouncing around at a high speed, even though the macro object of the "baseball" is standing still.
- The cool rapidly moving one is the opposite, it's constituent atoms are moving slowly, while the macro object is moving fast across space.
- The difference is in the manner of the motion, organized (thrown ball), disorganized (heat).

1.2 If the atoms of all objects are perpetually in motion, how can there be any permanent objects, such as fossil imprints?

- Atoms are attracted to one another (bonds with other atoms, electromagnetic force?), and that attraction holds them together until an external force rips them apart. They dont have enough energy alone to break those bonds. In solids that attraction causes them to form crystalline structures which are stable arrangements of atoms.
- That external force can be a chemical reaction, meaning it changes the composition of the atoms for instance creating molecules and ripping them from their original arrangement, or a non-chemical reaction like being struck by something.

1.3 Explain qualitatively why and how friction in a moving machine produces heat. Explain also, if you can, why heat cannot produce useful motion by the reverse process.

- Heat is nothing more than the movement of atoms, the more movement the more heat.
- When two solids are rubbing against each other, the (organized) mechanical energy going into the system becomes (disorganized) internal energy in the material and the surrounding environment.
- The microscoping imperfections of the solids are grinding against each other, catching, releasing, vibrating, and if we zoom in further that means all that chaos is pushing and pulling atoms around.
- The second part of the question has to do with entropy (second law of thermodynamics). A system can go from an ordered state to a disordered state but not the other way around. Thermodynamic processes are not reversible.

1.4 Chemists have found that the molecules of rubber consist of long criss-crossed chains of atoms. Explain why a rubber band becomes warm when it is stretched.

- High entropy at rest: long polymer chains are tangled.
- Low entropy when deformed: chains are more ordered.
- Temperature and entropy are inversely related.
- So as we stretch, the entropy drops, and thus temperature has to increase. And vice versa.

1.5 What should happen to a rubber band which is supporting a given weight, if it is heated? (To find out, try it.)

- You add heat $Q > 0$.
- This forces the molecules to move toward a higher entropy state (S¿0).
- The higher entropy state for a polymer is tangled and coiled.
- Coiled chains shorten the band, causing it to contract and lift the weight.

1.6 Can you explain why there are no crystals that have the shape of a regular pentagon? (Triangles, squares, and hexagons are common in crystal forms.)

- Because you can't pack pentagons on the Euclidean plane, as 2pi is not divisible by 108 degrees. So you can't tesselate a plane. (No translational symmetry atomic structure.

1.7 You are given a large number of steel balls of equal diameter d and a container of known volume V. Every dimension of the container is much greater than the diameter of a ball. What is the greatest number of balls, N, that can be placed in the container

- $4/3pir^3 = S$ (volume of sphere)
- However packing a 3d space with spheres will create gaps
- So what is the tesselation pattern of perfect spheres? We need to calculate the in between space
- N = V/(S)
- N = V/(S+X)
- Gauss's sphere packing rule

1.8 How should the pressure P of a gas vary with n, the number of atoms per unit volume, and (v), the average speed of an atom? (Should P be proportional to n and/ or to (v), or should it vary more, or less, rapidly than linearly?)

- P = pressure
- n = atoms/unit volume
- v = average speed
- average speed is heat so the more it heats up the greater the pressure regardless of density (n)
- P also goes up with more density regardless of velocity staying constant
- $P = 1/3N/Vmv^2$
- Linearly with more density, and quadratically with average speed

2 Chapter 2

Mathematically, the optimal path P^* is the one that minimizes the sum of the costs (or weights) of the edges it traverses:

$$P^* = \arg\min_{P} \sum_{i=1}^{n-1} w(v_i, v_{i+1})$$

where $w(v_i, v_{i+1})$ is the cost to travel between adjacent vertices v_i and v_{i+1}. This cost is typically a function of distance and proximity to known obstacles on the costmap.